# PromoLens: Production-Grade AI Microdrama Promo Extraction & Ranking

**Target Track: Problem 1 — Find the Promos (Ads) | Evaluation Dataset: 50 Screenplay Episodes (Season 1)**

## 1. Problem Choice, Strategic Framing & Core Assumptions

**Why Problem 1?** At Kuku, AI microdramas are acquired and monetized primarily through short vertical video ads on TikTok, Instagram Reels, and YouTube Shorts. Creative editing is today's primary operational bottleneck: human editors spend ~18 hours per series scrubbing 50+ episodes to find 20–60s hooks. Problem 1 directly impacts customer acquisition cost (CAC), viewer retention, and content velocity. It provides an objective, benchmarkable challenge where narrative psychology meets ad performance.

**Core Operational Assumptions:**

- **Strict Platform Duration (20s–60s):** <20s lacks narrative substance to establish conflict; >60s fails vertical ad platform guidelines.
- **Zero-Splice Boundary Alignment:** Cuts must strictly snap to natural dialogue utterance onsets and breath/punctuation terminations or audio stingers. Never cut an actor mid-word.
- **3-Second Hook Rule:** Cold TikTok scrolling requires immediate high entropy in the first 3s (visceral betrayal, public humiliation, slap, or status threat).
- **Zero-Lore Standalone Clarity:** A cold viewer at 1 AM must instantly identify who is wronged and what is at stake without backstory.
- **Zeigarnik Cliffhanger:** Abrupt cut precisely at the peak of unresolved emotional tension, compelling immediate app download.

## 2. System Architecture: Why a Hybrid Neuro-Symbolic Pipeline?

Brute-forcing 50 full scripts (~150,000 words) into an LLM produces hallucinated timestamps, broken sentence boundaries, high latency, and prohibitive API bills (~$8.50/show). Pure keyword regex lacks semantic understanding of unresolved cliffhangers. We built **PromoLens** as a modular, 5-stage hybrid architecture:

- **Stage 1: Deterministic Screenplay Parser:** Robust state machine parsing 50 episodes, extracting 100 scenes, 3,727 camera/action elements, 1,487 dialogue lines with parenthetical emotional cues (`Cold, Commanding`), and normalizing micro-timestamps (`MM:SS.ss`).
- **Stage 2: Constraint-Solver Window Generator:** Slides boundary-aligned temporal windows adhering strictly to [20.0s, 60.0s], evaluating 32,984 candidate permutations across the season in <1.1 seconds.
- **Stage 3: Multi-Dimensional Scoring Engine:** Evaluates each window via weighted vector scoring:
  **Composite Score = 0.35 · Hook + 0.30 · Cliffhanger + 0.20 · Standalone Clarity + 0.15 · Pacing**
- **Stage 4: Temporal Non-Maximum Suppression (NMS):** Implements Intersection over Union (IoU ≥ 0.30) to eliminate redundant overlapping cuts of identical scenes, ensuring season-wide promo diversity.
- **Stage 5: Swappable Model Bridge:** Runs 100% offline via our calibrated heuristic scorer, while exposing an swappable interface for Gemini 1.5 Flash / GPT-4o-mini if an interviewer provides an API key.

## 3. Real-World Scale, Unit Economics & Human-in-the-Loop

| Evaluation Dimension | Brute-Force LLM Approach | PromoLens Hybrid Engine |
|---|---|---|
| Execution Time (50 Episodes) | ~12 to 15 minutes | **1.07 seconds (0.02s / episode)** |
| Compute / API Cost per Show | ~$8.50 per 50-ep series | **$0.00 (Local) / $0.08 (Pruned LLM)** |
| Monthly Spend (100 Shows) | ~$850.00 / month | **<$8.00 / month (>99% reduction)** |
| Boundary Alignment Precision | 68% (frequent mid-word cuts) | **100% Exact Math Alignment** |
| Production Asset Output | Raw unstructured text | **JSON, CSV, and Premiere Pro EDLs** |

**Human-in-the-Loop (HITL) Workflow:** Video editors never scrub blind. PromoLens generates instant CMX 3600 Edit Decision Lists (EDLs) imported directly into Adobe Premiere Pro or DaVinci Resolve. Editors review only the top-ranked candidates, reducing promo delivery time from 18 hours to under 45 minutes per show.

## 4. Strategic Ranking: Which Promos to Post First & Why?

Ad creative performance depends entirely on platform audience psychology and ad fatigue cycling. Promos are ranked into 3 tactical funnels:

| Posting Wave & Funnel | Top Extracted Promo & Timestamps | Why Post First? (Psychological Rationale) |
|---|---|---|
| **Wave 1: Cold User Acquisition**<br>(Top-of-Funnel / Post FIRST)<br>*TikTok & Meta Broad Reels* | **Rank #1: Ep 2 [00:53.40 – 01:16.40] (23.0s)**<br>**Hook:** *'You look like a kitchen maid who wandered into the wrong wedding.'*<br>**Cliffhanger:** Juliana turns with fury: *'Dignity?'*<br>**Rank #2: Ep 2 [00:00.00 – 00:27.40] (27.4s)**<br>**Hook:** Gerald dumps bride at altar: *'I cannot marry you.'* | Cold scrolling viewers have **zero affinity with lore**. Only high-contrast moral outrage, public humiliation, and cruel betrayal trigger thumb-stops in <2s. High completion rate without prior exposition. |
| **Wave 2: Emotional Escalation**<br>(Mid-Funnel / Post SECOND)<br>*Meta Video & In-App Ads* | **Rank #3: Ep 30 [01:14.80 – 01:48.80] (34.0s)**<br>**Hook:** *'Give those facts to a magistrate in order, and she becomes a poisoner.'*<br>**Cliffhanger:** Catherine demands: *'And the Dowager?'*<br>**Rank #5: Ep 26 [00:55.80 – 01:35.00] (39.2s)**<br>**Hook:** *'Rupert & Catherine are poisoning me in bed.'* | Retargets viewers who engaged with Wave 1. Introduces life-or-death stakes, murder framing, and protective romance. Drives consideration and series bingeing into direct app installs. |
| **Wave 3: Cathartic Vindication**<br>(Retargeting / Post THIRD)<br>*YouTube Shorts & Coin Unlocks* | **Episode 42 [00:17.90 – 00:54.00] (36.1s)**<br>**Hook:** Corrupt lord yells: *'Guards! Arrest this imposter!'*<br>**Cliffhanger:** The broken soldier emerges in full ducal regalia, and royal guards salute him. | Re-activates stalled or churned users. Delivers the ultimate revenge payoff (the broken beggar is secretly the Grand Duke), compelling users to unlock paid episodes. |

## 5. Quality Checking, Error Auditing & Honest Failure Analysis

We ran an automated diagnostic test suite (`src/quality_checker.py`) across all generated promos. Of 25 top promos, **17 passed clean for immediate render**, 3 were flagged for context review, and 5 required cut boundary tuning. Here is where the tool picks badly and our mitigations:

• **1. Pronoun Antecedent Ambiguity (Cold-Viewer Confusion):**
*Failure Mode:* In Ep 32 [00:47.80–01:19.40], opening line is: *'A disgraced woman? Your soldier?'* Without prior context, cold viewers do not know who 'your soldier' refers to.
*Remedy:* Built `PRONOUN_AMBIGUITY` detector that flags cuts opening on unresolved 3rd-person pronouns, routing them to editorial review.
• **2. Non-Verbal Tension Blind Spot (Visual Action vs Low Text Density):**
*Failure Mode:* In Ep 1 [02:11.30–02:17.00], Juliana's knuckles whiten and she refuses to weep as dark strings swell. Pure NLP models score this low due to zero dialogue.
*Remedy:* Weighted `SOUND FX` ('cliffhanger drone swells') and `ACTION` ('knuckles whiten') to reward non-verbal visual acting.
• **3. False Cliffhangers (Cut on Declarative Statements):**
*Failure Mode:* Dialogue ending on flat periods (e.g., *'You have a solution.'*) sounds like an ordinary pause rather than an unresolved cliffhanger.
*Remedy:* Implemented interrogative boost (`?`), penalizing non-interrogative endings unless paired with sharp audio stingers or physical climaxes.
• **Tool Reflection (Where AI Failed Us):** Early zero-shot prompts to LLMs produced hallucinated cutoffs (e.g. cutting Juliana off at 'aren't t'). We completely eliminated this flaw by constraining candidate generation to deterministic script element boundaries.

## 6. Deliverables & Prototype Verification

• **Interactive Streamlit Studio:** `streamlit run app.py` — Live timeline scrub, filter by campaign wave, test custom scripts.
• **One-Command Production CLI:** `python run_pipeline.py` — Processes all 50 episodes in 1.07s; exports JSON, CSV, and Premiere EDLs.
• **Raw Outputs Folder:** Complete unedited dataset results in `raw_outputs/` (includes `all_promos.json`, `top_ranked_promos.csv`, `quality_diagnostics.json`, and 10 NLE EDLs).